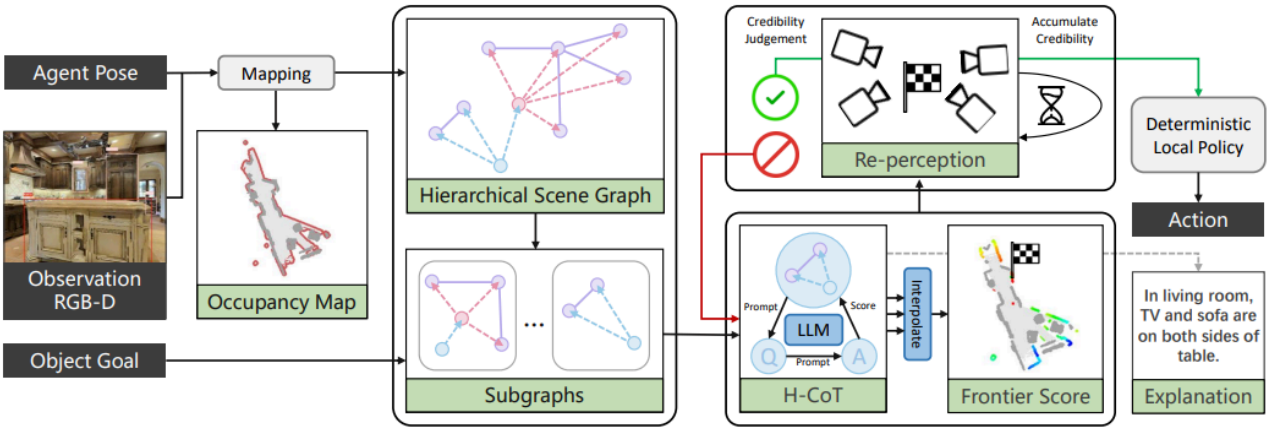
SG-Nav: Online 3D Scene Graph Prompting for LLM-based Zero-shot Object Navigation

> #Navigation #EmbodiedAI



- What problems does this framework solve?
  - With pure text, LLMs cannot do deep spatial logical inference
  - It is unreadable and unreliable to let the model directly output the confidence score of the next edge to be explored.
  - Lack of ability to correct mistakes.
- What does this framework do?
  - Scene Construction
    - There are 3 types of nodes: object (a table), group (the chairs around that table), and room.
    - There are 2 types of edges: affiliation (the table is in the dining hall) and relationship (the cup is on the table).
    - Realtime graph construction: Graph construction can be time-costly, so to avoid high latency, the paper proposes a way of incremental construction. Traditionally, when there are m new nodes and n old nodes, the worst performance can be $\mathcal{O}(m(m+n))$. To avoid this, the paper proposes a prompt to let the LLM predict all relationships of all nodes in one run, decreasing the complexity to $\mathcal{O}(m)$.
    - Pruning: Short edges (i.e., close nodes) will be validated by VLM's, and long edges will be validated by physical conditions (i.e., in the same room and can be seen together). Edges that don't fulfill the condition will be pruned.
  - Hierarchical Chain-of-Thought Prompting
    - The framework will split the scene into several sub-scenes and run the following CoT to predict:
      - The distance between the center objects and the target.
      - Ask questions about the object and target.
      - Based on the sub-scenes, answer the questions.
      - Summary and output a probability that the target is in the sub-scene P^{sub}.

        - The frontier's score (to decide the next exploration) is given by:

          $P_i^{fro} = \sum_{j=1}^{M} \frac{P_j^{sub}}{D_{ij}}$

            - M is the # of sub-scenes.
            - i is the i-th unexplored frontier.
            - D_{ij} is the distance between the sub-scene center and the frontier.
    - To avoid the false-positive, after the agent finds a suspect target, it will not stop. Instead, it will approach the object, and observe from multiple POVs. The score is given by:

      $S_k = C_k \cdot \sum_{j=1}^{M} \frac{P_j^{sub}}{D_j}$

        - C_k is the raw score given by the visual module.
        - P_j^{sub} is the probability that the target is in the sub-scene.
        - D_j is the distance between the suspect object and the center of j-th sub-scene.
    - Finally, if the score is higher than the threshold, the navigation will be announced as a success.

      $N_{stop} < N_{max} \text{ where } N_{stop} = N \text{ s.t. } \sum_{i=1}^{N-1} S_i < S_{thres} \leq \sum_{i=1}^{N} S_i$
- Experiment Results
    - Significantly outperforms SOTA: SG-Nav achieves a success rate (SR) that exceeds that of previous state-of-the-art zero-shot methods by more than 10% across all benchmarks.
    - A landmark breakthrough: On the highly challenging MP3D dataset, SG-Nav is the first zero-shot navigation framework to match or even surpass the performance of supervised learning methods.
    - Module ablation validates effectiveness: Ablation experiments show that without the re-perception mechanism (RP), the agent suffers a significant performance drop due to false positive errors; similarly, removing the subgraph design leads to a precipitous decline in SR, fully demonstrating the necessity of the scene graph and error correction mechanisms.
- Work Cited
    - Paper
    - Chat History
    - NotebookLM